

Grant Application: Φ -Hook

A Parameter-Free, Mathematically Immutable LP Retention Layer for Uniswap V4

To: Uniswap Foundation Grants Committee

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Date: May 17, 2026

Amount Requested: \$40,000 USD | Duration: 12 Weeks

1. The Problem

Uniswap V4 unlocks unprecedented customization, yet it inherits DeFi's oldest vulnerability: mercenary liquidity. Today's retention strategies rely on two fragile crutches—inflationary governance tokens and manually tuned fee parameters—that demand constant committee oversight and inevitably decay into extraction games for sophisticated actors.

2. The Solution

Φ -Hook (Phi Hook) is a fully autonomous, zero-governance hook infrastructure that replaces every discretionary parameter with mathematical constants. By anchoring commitment tiers, reward compounding, and protocol fees to the Golden Ratio ($\phi \approx 1.618$) and Fibonacci sequences, Φ -Hook creates a self-regulating retention curve that deepens organically with time, not with emissions.

There are no admin keys to exploit, no fee dials to lobby, and no inflationary supply schedules. The protocol fee is permanently hardcoded to the Fibonacci retracement of 38.2% (3,820 / 10,000), streaming real yield directly to long-term LPs while penalizing early exits through deterministic, on-chain time-gates.

3. Current Status: Production-Ready

This is not a theoretical proposal. The core system is:

- 100% complete and compiled under Foundry
- Fully functional on the Sepolia testnet
- Verified with 89 passing functional and invariant vectors covering precision-loss edge cases, access-control restrictions, and dust-recycling mechanics
- Built with a modular, zero-dependency architecture ready for security review

4. Grant Request & Roadmap

We are requesting \$40,000 USD over 12 weeks to transition from testnet validation to production-grade infrastructure. Funds will be allocated exclusively to engineering and security:

Phase	Timeline	Deliverable
Phase 1	Weeks 1-3	Stateful invariant fuzzing expansion (Echidna + Foundry) with formalized property specs
Phase 2	Weeks 4-6	Independent security audit by a recognized Web3 security firm
Phase 3	Weeks 7-9	TypeScript SDK + Subgraph architecture for position tracking and reward forecasting
Phase 4	Weeks 10-12	Production LP dashboard deployment and mainnet launch coordination

5. Why Uniswap V4 Needs This

V4's hook architecture was designed to enable opinionated pool behavior without forking the core protocol. Φ -Hook is the purest expression of that philosophy: an opinionated, mathematically immutable retention layer that any pool deployer can permissionlessly integrate. It transforms V4 from a customizable AMM into a self-sustaining liquidity protocol.

6. Repository Access

The repository is held privately during the evaluation window to protect the design architecture. We will grant immediate read-only access to your technical review team upon request, including full test traces and Sepolia deployment addresses.

7. Closing

Thank you for your time and consideration. We are prepared to begin Phase 1 immediately upon approval and are committed to delivering a security-audited, fully documented hook to the Uniswap V4 ecosystem.

Sincerely,
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